

PROBLEM SET 6

Due April 26.

1. Consider a symmetric independent private values setting with two potential buyers. The seller owns an indivisible object which she is commonly known to value at 0. Suppose that $v_i \in \{1, 2, 3\}$ for $i = 1, 2$ and that each realization is equally likely (occurs with probability $1/3$). Suppose also that $S_i = \{No, 1, 2, 3\}$ for all $i = 1, 2$ so that there are only four possible actions for each bidder. Suppose that ties are broken randomly so that each bidder acquires the object with probability $1/2$.
 - (a) Suppose that the seller announces a second-price sealed-bid auction with reserve price 1. Find all symmetric BNE $b^*(v)$, in which the bidders play weakly dominant strategies. How much expected revenue is generated in this BNE?
 - (b) Suppose that the seller announces a second-price sealed bid auction with reserve price 1. Find an asymmetric BNE $(b_1(v_1), b_2(v_2))$.
 - (c) Suppose that the seller announces a first-price sealed bid auction with reserve price of 1. Find the symmetric BNE, $b^*(v)$ which generates the highest expected revenue for the seller. How much expected revenue is generated in this BNE?
 - (d) Suppose the seller announces an all-pay auction with reserve price of 1, in which the bidder pays his bid independent of the auction outcome. Find a symmetric BNE, $b^{***}(v)$. How much expected revenue is generated in this BNE?
2. Consider a symmetric independent private values setting in which there are $n > 1$ buyers whose valuations are independently drawn from a uniform distribution on $[0, 1]$. Consider an all-pay auction with a reserve price and no entry fee. Note: in an all-pay auction, all bidders pay their bids, but the bidder with the highest bid wins the object.
 - (a) Compute the optimal reserve price for the seller r^* .
 - (b) Given r^* , find the symmetric BNE bidding function $b^*(v)$.
 - (c) Calculate the seller's expected equilibrium revenue.
 - (d) Is this an optimal sales mechanism from the perspective of the seller? Explain.
3. (Qualifying exam 2011) Consider a seller (s) who wishes to sell an item to one of two potential buyers ($i = 1, 2$). Each buyer's valuation v_i is identically and independently distributed according to a uniform distribution with support $v_i \in [0, 1]$ and the seller's valuation is known to be $v_s = 0$. The seller employs an auction consisting of two stages. In the first stage, both buyers choose simultaneously whether to pay the participation fee c for the right to participate in the auction. The buyers who have paid the participation fee are then given access to the room in which the auction is conducted. Thus, each buyer is able to observe whether the competing buyer has chosen to enter the auction. The second stage of the auction is a simple first price auction with 0 reserve price, in which the highest bidder wins the auction and pays his own bid.
 - (a) Suppose that bidder i has observed that the competing bidder has chosen to stay out of the auction. Find the bid $b(v_i|out)$ that is part of BPE.
 - (b) Suppose that the bidders follow a symmetric entry strategy. Bidder i enters the auction only if $v_i \geq v_0$ and stays out if $v_i < v_0$. Consider the second stage, in which bidder i observes that both him and the competing seller have entered the auction. Let $\tilde{F}(\cdot)$ denote the cumulative distribution of v conditional on $v \geq v_0$. Find $\tilde{F}(v)$.
 - (c) Suppose that the bidders use a symmetric bid function $b(v|in)$ that is increasing in the bidder's valuation. Derive $b(v|in)$.
 - (d) What is the economic intuition behind the bidding function $b(v|in)$ that you derived in part c). Find v_0 as a function of the participation cost c .
 - (e) Find the seller's expected revenue from the auction $\Pi_s(c)$ as a function of c .
 - (f) What is the optimal participation cost c^* from the point of view of the seller?